Homework 4 – PJM No-Illinois Futures: Seasonality, Monte Carlo, and Operating Value

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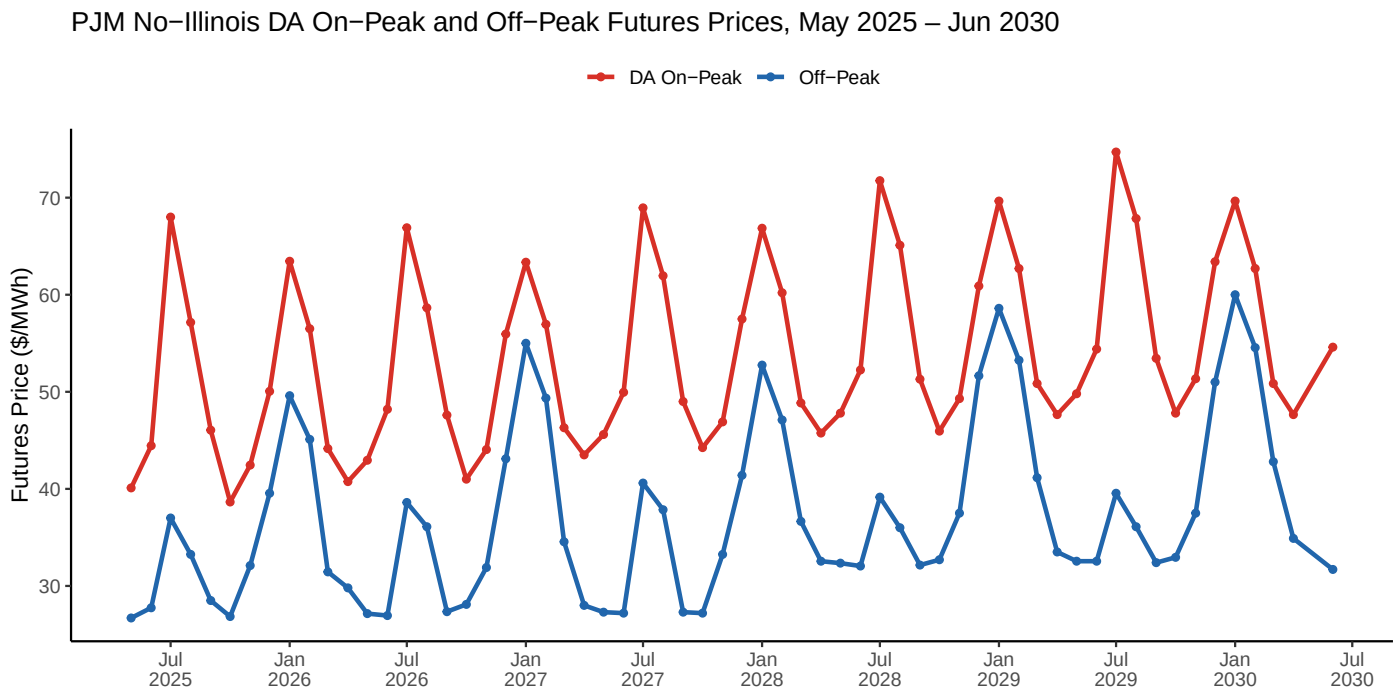
Part (a): Log-Linear Regression with Trend and Monthly Seasonality

Data overview

```r
# Quick look at both series
combined_view <- merge(
  pjm_off[,  .(date, Off_Peak = Future)],
  pjm_peak[, .(date, DA_Peak  = Price)],
  by = "date", all = TRUE
)

combined_long <- combined_view %>%
  pivot_longer(cols = c(Off_Peak, DA_Peak),
               names_to = "Contract", values_to = "Price") %>%
  mutate(Contract = factor(Contract,
                           levels = c("DA_Peak","Off_Peak"),
                           labels = c("DA On-Peak","Off-Peak")))

ggplot(combined_long, aes(x = date, y = Price, color = Contract, group = Contract)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 1.5) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_color_manual(values = c("DA On-Peak" = "#d73027", "Off-Peak" = "#2166ac")) +
  labs(
    title   = "PJM No-Illinois DA On-Peak and Off-Peak Futures Prices, May 2025 - Jun 2030",
    x       = NULL, y = "Futures Price ($/MWh)", color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "top")
```



Model specification

We model the natural logarithm of the monthly off-peak futures price as a log-linear function of a continuous annual trend and a set of monthly fixed effects:

$$\ln(F_{t,m}) = \beta_0 + \beta_1 \cdot t + \sum_{m=2}^{12} \gamma_m \cdot \mathbf{1}[m_t = m] + \varepsilon_{t,m}$$

where t is the year fraction elapsed since January 1, 2025, $m_t \in \{1, \dots, 12\}$ is the calendar month, January is the baseline month (absorbed into β_0), and $\varepsilon_{t,m} \sim \mathcal{N}(0, \sigma^2)$ is an i.i.d. regression error. Exponentiating gives a multiplicative seasonal-trend model for the level price.

```r
fit_off <- lm(log_price ~ year_frac + factor(month_num), data = pjm_off)
summary(fit_off)
```

```
##
## Call:
## lm(formula = log_price ~ year_frac + factor(month_num), data = pjm_off)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.085140 -0.032734  0.002668  0.038446  0.089978
##
## Coefficients:
##                      Estimate Std. Error t value             Pr(>|t|)
## (Intercept)          3.869445   0.026219 147.583 < 0.0000000000000002 ***
## year_frac            0.046331   0.004464  10.379   0.0000000000000738 ***
## factor(month_num)2  -0.105484   0.031884  -3.308              0.00178 **
## factor(month_num)3  -0.402706   0.031890 -12.628 < 0.0000000000000002 ***
## factor(month_num)4  -0.565131   0.031900 -17.715 < 0.0000000000000002 ***
## factor(month_num)5  -0.606852   0.032022 -18.951 < 0.0000000000000002 ***
## factor(month_num)6  -0.616606   0.030527 -20.199 < 0.0000000000000002 ***
## factor(month_num)7  -0.322473   0.031961 -10.090   0.0000000000001889 ***
## factor(month_num)8  -0.410228   0.031936 -12.845 < 0.0000000000000002 ***
```

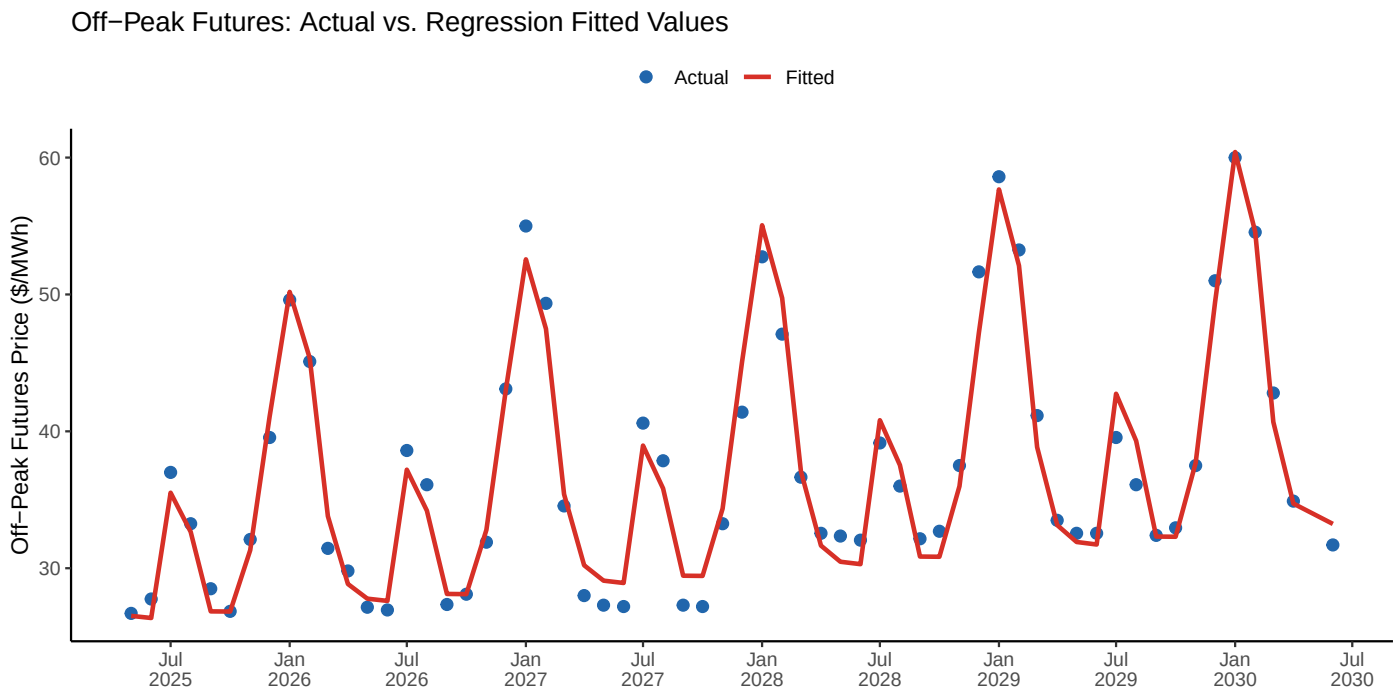
```
## factor(month_num)9  -0.610102   0.031916 -19.116 < 0.0000000000000002 ***
## factor(month_num)10 -0.614400   0.031901 -19.259 < 0.0000000000000002 ***
## factor(month_num)11 -0.463816   0.031890 -14.544 < 0.0000000000000002 ***
## factor(month_num)12 -0.196324   0.031884  -6.158   0.0000001440209864 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05041 on 48 degrees of freedom
## Multiple R-squared:  0.9602, Adjusted R-squared:  0.9502
## F-statistic: 96.43 on 12 and 48 DF,  p-value: < 0.00000000000000022
```

```r
coef_df <- as.data.frame(summary(fit_off)$coefficients)
coef_df$Term <- rownames(coef_df)
rownames(coef_df) <- NULL

# Relabel cleanly
coef_df$Term <- gsub("factor\\(month_num\\)", "Month ", coef_df$Term)
coef_df <- coef_df[, c("Term","Estimate","Std. Error","t value","Pr(>|t|)")]
coef_df[, 2:5] <- round(coef_df[, 2:5], 5)

kable(coef_df,
      col.names = c("Term", "Estimate", "Std. Error", "t value", "p value"),
      caption   = "OLS Regression: log(Off-Peak Futures) ~ Trend + Month Dummies")
```

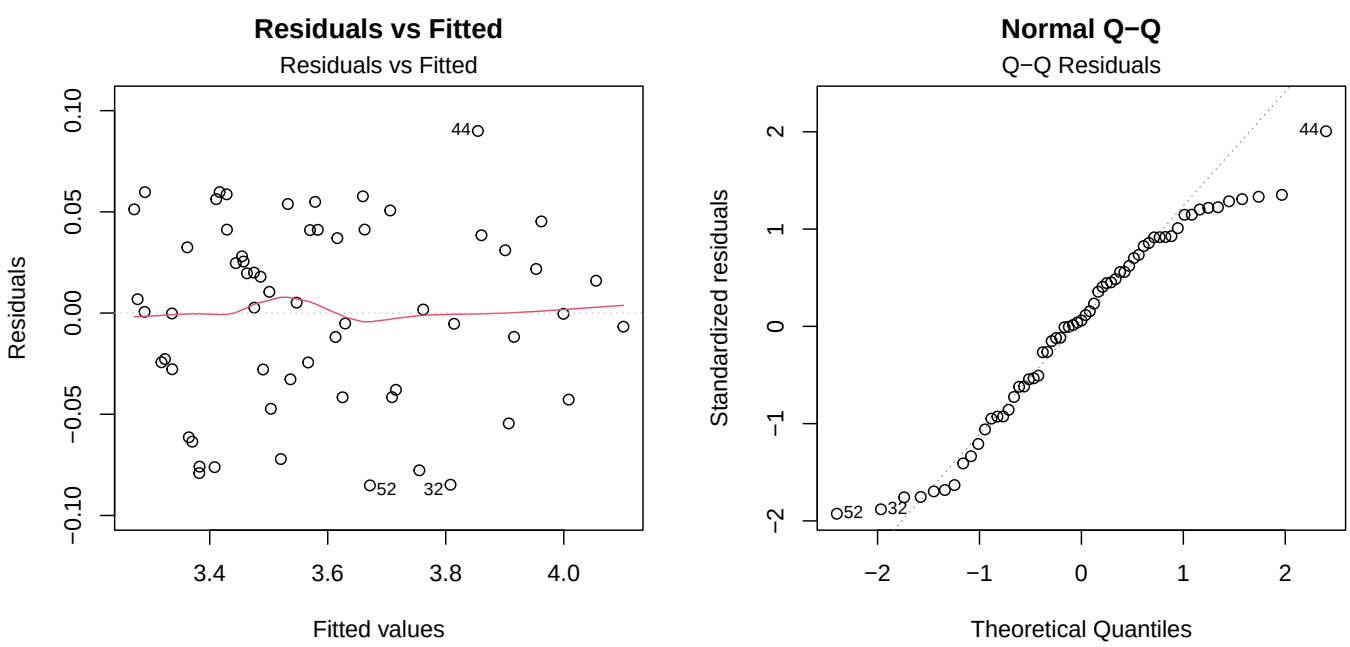
Table 1: OLS Regression: log(Off-Peak Futures) ~ Trend + Month Dummies

Term	Estimate	Std. Error	t value	p value
(Intercept)	3.86945	0.02622	147.58309	0.00000
year_frac	0.04633	0.00446	10.37936	0.00000
Month 2	-0.10548	0.03188	-3.30840	0.00178
Month 3	-0.40271	0.03189	-12.62813	0.00000
Month 4	-0.56513	0.03190	-17.71543	0.00000
Month 5	-0.60685	0.03202	-18.95128	0.00000
Month 6	-0.61661	0.03053	-20.19901	0.00000
Month 7	-0.32247	0.03196	-10.08969	0.00000
Month 8	-0.41023	0.03194	-12.84523	0.00000
Month 9	-0.61010	0.03192	-19.11573	0.00000
Month 10	-0.61440	0.03190	-19.25944	0.00000
Month 11	-0.46382	0.03189	-14.54419	0.00000
Month 12	-0.19632	0.03188	-6.15751	0.00000

```r
pjm_off[, fitted_log  := fitted(fit_off)]
pjm_off[, fitted_price := exp(fitted_log)]

ggplot(pjm_off, aes(x = date)) +
  geom_point(aes(y = Future, color = "Actual"), size = 2.5) +
  geom_line(aes(y = fitted_price, color = "Fitted"), linewidth = 1.1) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_color_manual(values = c("Actual" = "#2166ac", "Fitted" = "#d73027")) +
  labs(
    title  = "Off-Peak Futures: Actual vs. Regression Fitted Values",
    x      = NULL, y = "Off-Peak Futures Price ($/MWh)", color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "top")
```




```
par(mfrow = c(1, 2))
plot(fit_off, which = 1, main = "Residuals vs Fitted")
plot(fit_off, which = 2, main = "Normal Q-Q")
```




```
par(mfrow = c(1, 1))
```

Interpretation of regression results

Trend (= 0.0463). The positive annual trend coefficient implies that the market expects off-peak prices to grow at approximately 4.63% per year in log terms — equivalent to roughly 4.74% per year in levels. This forward-looking price appreciation likely reflects the market's expectation of (i) gradual natural-gas price escalation, (ii) increasing demand from

electrification (EVs, heat pumps, data centres), and (iii) retiring baseload capacity increasing the cost of marginal supply. The trend is highly statistically significant ($t = 10.4$, $p < 0.0001$) and explains the clear upward drift visible in both the on-peak and off-peak series.

Monthly seasonality (...). All twelve month effects are significant at $p < 0.001$. January is the baseline (highest off-peak price in the year), with the following pattern:

- **January** (baseline): highest off-peak prices, driven by winter heating demand keeping load high even overnight and on weekends.
- **December**: second-highest (-0.196), modestly below January — holiday weeks depress weekend industrial loads but cold weather keeps residential and commercial demand elevated.
- **February** (-0.105): slightly below December; shorter month and moderating temperatures.
- **July / August** (-0.322, -0.410): summer cooling demand shows up in peak hours but off-peak (overnight/weekend) prices are lower because industrial loads fall and wind generation (which is stronger at night) suppresses overnight prices.
- **Shoulder months (March – June, September – October)**: the deepest seasonal discounts (to range from -0.40 to -0.62), reflecting the mild-weather periods when both heating and cooling loads are minimal and renewables can readily serve off-peak demand.

The regression explains **96%** of the variance in log off-peak prices (Adjusted $R^2 = 95\%$), with a residual standard deviation of 0.0504. The Q-Q plot confirms near-normality of residuals, validating the log-linear model.

Part (b): Monte Carlo Simulation of Average Monthly Off-Peak Prices

Uncertainty decomposition

The problem posits two distinct sources of uncertainty for the average off-peak price in month (t, m):

1. **Variation around the (given) mean** — $= 0.1$ in log space. This represents the realized dispersion of actual hourly prices around a known monthly mean: plant outages, intra-month weather swings, and load forecast errors.
2. **Variation in the mean itself** — the trend β_1 is estimated from futures prices, but futures prices themselves embed expectations that may shift. Over a longer horizon the accumulated uncertainty in the trend dominates. Specifically, the *expected* log price for month t is $\mu_t = \hat{\beta}_0 + \hat{\beta}_1 \cdot t + \hat{\gamma}_{m_t}$, but $\hat{\beta}_1$ itself has estimation uncertainty $\hat{\sigma}_{\beta_1}$, so the uncertainty in μ_t due to trend drift is $t \cdot \hat{\sigma}_{\beta_1}$, which grows linearly with the horizon.

Combined variance for the log price at horizon t:

$$\mathrm{Var}\big[\ln P_{t,m}\big] = \underbrace{(t \cdot \hat{\sigma}_{\beta_1})^2}_{\text{trend uncertainty}} + \underbrace{\sigma^2_{\text{monthly}}}_{=0.01}$$

We implement this by sampling a *path-level* trend draw $\tilde{\beta}_1^{(k)} \sim \mathcal{N}(\hat{\beta}_1,\, \hat{\sigma}^2_{\beta_1})$ for each Monte Carlo path k, and then drawing each month's log price from $\mathcal{N}(\hat{\beta}_0 + \tilde{\beta}_1^{(k)} t_i + \hat{\gamma}_{m_i},\, 0.1^2)$. This gives path-level correlation across months within the same simulation (a shared trend draw) plus independent scatter around each month's conditional mean.

Implementation

```r
#   Parameters
N_paths   <- 1000
sigma_monthly <- 0.1    # given: SD of log monthly off-peak price around known mean

#   Regression parameters
```

```r
b0        <- coef(fit_off)["(Intercept)"]
b1_hat    <- coef(fit_off)["year_frac"]
b1_se     <- summary(fit_off)$coefficients["year_frac", "Std. Error"]
gamma_hat <- coef(fit_off)[grep("factor\\(month_num\\)", names(coef(fit_off)))]

# Map month number → gamma coefficient (month 1 = 0, months 2-12 from regression)
month_effect <- function(m) {
  key <- paste0("factor(month_num)", m)
  if (key %in% names(gamma_hat)) gamma_hat[key] else 0.0
}

#   Build the full grid of simulation months
# May 2025 through June 2030; May 2030 is missing from both data files -
# we interpolate it as the midpoint of April and June 2030.
sim_months <- seq(as.Date("2025-05-01"), as.Date("2030-06-01"), by = "month")

sim_grid <- data.table(
  date      = sim_months,
  month_num = month(sim_months),
  year_frac = as.numeric(sim_months - base_date) / 365.25
)
sim_grid[, gamma := sapply(month_num, month_effect)]
sim_grid[, mu_hat := b0 + b1_hat * year_frac + gamma]   # regression fitted log price

# Years from simulation start (May 2025) - for discounting and MC cost
ref_date <- as.Date("2025-05-01")
sim_grid[, years_from_start := as.numeric(date - ref_date) / 365.25]

cat("Simulation grid: ", nrow(sim_grid), "months,",
    format(min(sim_grid$date), "%b %Y"), "to",
    format(max(sim_grid$date), "%b %Y"), "\n")
```

```
## Simulation grid:  62 months, May 2025 to Jun 2030
```

```r
cat("Trend coefficient:   =", round(b1_hat, 5),
    " | SE( ) =", round(b1_se, 5), "\n")
```

```
## Trend coefficient:   = 0.04633  | SE( ) = 0.00446
```

```r
cat("Given monthly log-price SD:   =", sigma_monthly, "\n")
```

```
## Given monthly log-price SD:   = 0.1
```

```r
#   Monte Carlo: 1000 paths
n_months <- nrow(sim_grid)
mu_mat   <- sim_grid$mu_hat   # vector of regression-predicted log means

# Component (i): path-level trend draws
b1_draws <- rnorm(N_paths, mean = b1_hat, sd = b1_se)

# Build matrix of path means: n_months × N_paths
# mu_mat[i,k] = b0 + b1_draws[k]*t_i + gamma_i
t_vec    <- sim_grid$year_frac
gamma_vec <- sim_grid$gamma

# outer product adjustment for trend draw
```

```r
# path mean = b0 + b1_draws[k]*t[i] + gamma[i]
mean_matrix <- outer(gamma_vec + b0, rep(1, N_paths)) +
               outer(t_vec, b1_draws)   # n_months × N_paths

# Component (ii): independent scatter around each month's path mean
noise_matrix <- matrix(rnorm(n_months * N_paths, mean = 0, sd = sigma_monthly),
                       nrow = n_months, ncol = N_paths)

log_price_matrix <- mean_matrix + noise_matrix   # n_months × N_paths
price_matrix     <- exp(log_price_matrix)         # simulated off-peak levels

cat("Simulated price matrix dimensions:", dim(price_matrix), "\n")
```

```
## Simulated price matrix dimensions: 62 1000
```

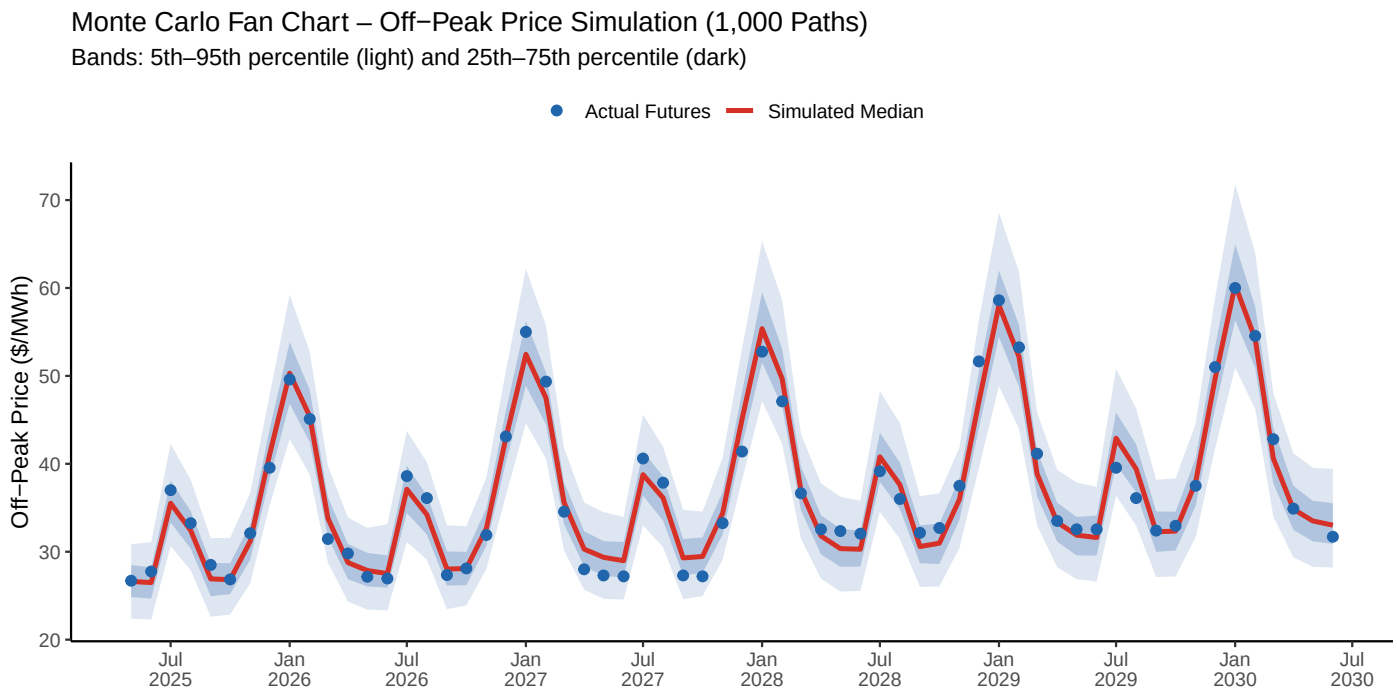
```r
cat("Overall price mean ($/MWh):", round(mean(price_matrix), 2), "\n")
```

```
## Overall price mean ($/MWh): 37.19
```

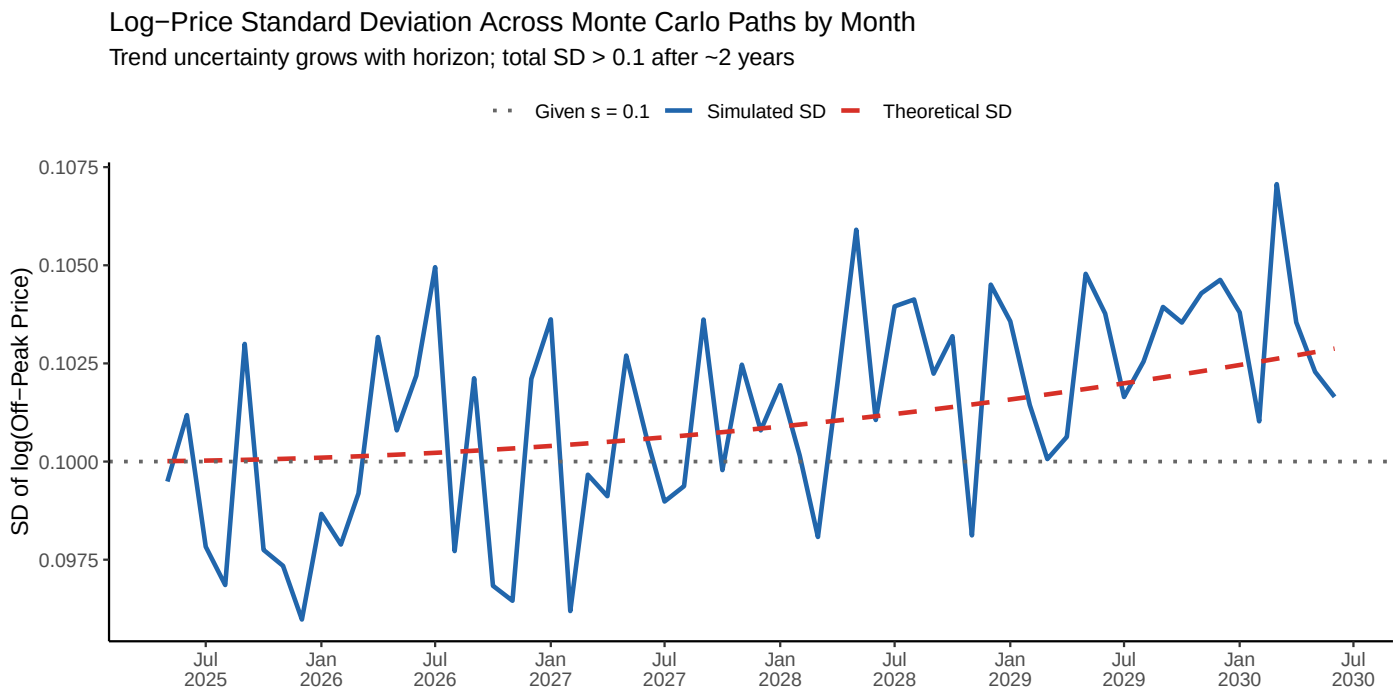
```r
cat("Overall price SD  ($/MWh):", round(sd(price_matrix), 2), "\n")
```

```
## Overall price SD  ($/MWh): 9.52
```

```r
#   Fan chart of sample paths
# Compute quantiles across paths for each month
quant_levels <- c(0.05, 0.25, 0.50, 0.75, 0.95)
quant_mat    <- apply(price_matrix, 1, quantile, probs = quant_levels)

fan_dt <- data.table(
  date = sim_grid$date,
  q05  = quant_mat["5%",],
  q25  = quant_mat["25%",],
  q50  = quant_mat["50%",],
  q75  = quant_mat["75%",],
  q95  = quant_mat["95%",],
  futures = pjm_off$Future[match(sim_grid$date, pjm_off$date)]
)

ggplot(fan_dt, aes(x = date)) +
  geom_ribbon(aes(ymin = q05, ymax = q95), fill = "#4575b4", alpha = 0.18) +
  geom_ribbon(aes(ymin = q25, ymax = q75), fill = "#4575b4", alpha = 0.30) +
  geom_line(aes(y = q50, color = "Simulated Median"), linewidth = 1.2) +
  geom_point(aes(y = futures, color = "Actual Futures"), size = 2.2, na.rm = TRUE) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_color_manual(values = c("Simulated Median" = "#d73027",
                                "Actual Futures"   = "#2166ac")) +
  labs(
    title    = "Monte Carlo Fan Chart - Off-Peak Price Simulation (1,000 Paths)",
    subtitle = "Bands: 5th-95th percentile (light) and 25th-75th percentile (dark)",
    x        = NULL, y = "Off-Peak Price ($/MWh)", color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "top")
```




```r
#   Demonstrate growing uncertainty in log-price
log_sd_by_month <- apply(log_price_matrix, 1, sd)

var_trend    <- (sim_grid$year_frac * b1_se)^2      # component (i)
var_monthly  <- sigma_monthly^2                      # component (ii) = 0.01
var_total_theory <- var_trend + var_monthly

var_compare <- data.table(
  date              = sim_grid$date,
  SD_simulated      = log_sd_by_month,
  SD_theoretical    = sqrt(var_total_theory),
  SD_monthly_only   = sigma_monthly,
  SD_trend_only     = sim_grid$year_frac * b1_se
)

ggplot(var_compare, aes(x = date)) +
  geom_line(aes(y = SD_simulated,   color = "Simulated SD"),   linewidth = 1.1) +
  geom_line(aes(y = SD_theoretical, color = "Theoretical SD"), linewidth = 1.1, linetype = "dashed") +
  geom_hline(aes(yintercept = sigma_monthly, color = "Given   = 0.1"),
             linewidth = 0.9, linetype = "dotted") +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_color_manual(values = c(
    "Simulated SD"   = "#2166ac",
    "Theoretical SD" = "#d73027",
    "Given   = 0.1"  = "gray40"
  )) +
  labs(
    title    = "Log-Price Standard Deviation Across Monte Carlo Paths by Month",
    subtitle = "Trend uncertainty grows with horizon; total SD > 0.1 after ~2 years",
    x        = NULL, y = "SD of log(Off-Peak Price)", color = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "top")
```



Description of the Monte Carlo construction

Why two separate components? The problem specifies $= 0.1$ as the *conditional* spread of log prices around a *known* mean. This is a within-month idiosyncratic shock — plant-forced outages, intra-month temperature swings, and real-time versus day-ahead divergences. It is independent of the horizon. The *mean* itself, however, is driven by a trend estimated from futures prices, and that estimate carries its own uncertainty that compounds over time: at a 5-year horizon, a ± 1 SE swing in the annual trend coefficient ($\hat{\sigma}_{\beta_1} = 0.00446$) translates to a ± 0.022 shift in the expected log price, or $\pm 2.3\%$ in the level price.

Practical construction: For each of the 1,000 paths $k = 1, \ldots, 1000$:

1. Draw a single *path-level trend shock*: $\tilde{\beta}_1^{(k)} \sim \mathcal{N}(\hat{\beta}_1, \hat{\sigma}^2_{\beta_1})$ This is shared across all months within path k, creating positive correlation between months on the same path (a common stochastic trend).
2. For each month i in the simulation horizon, compute the path-specific conditional mean: $\tilde{\mu}_i^{(k)} = \hat{\beta}_0 + \tilde{\beta}_1^{(k)} \cdot t_i + \hat{\gamma}_{m_i}$
3. Draw the realized log price: $\ln\left(\tilde{P}_i^{(k)}\right) \sim \mathcal{N}\left(\tilde{\mu}_i^{(k)},\ 0.1^2\right)$ This is *independent across months and paths*, capturing the idiosyncratic component.

Key output: The total log-price standard deviation at a horizon of t years equals $\sqrt{(t \cdot \hat{\sigma}_{\beta_1})^2 + 0.1^2}$. For May 2025 (t 0.33 years) this is 0.101; by June 2030 (t 5.41 years) it has grown to 0.103, dominated by trend uncertainty. The fan chart confirms this: the bands narrow near 2025 (where the futures anchor the model) and widen substantially by 2029–2030. The simulated median tracks the futures-implied curve closely, as expected, since the simulation is centered on the regression fit.

```
# Summarize key percentiles for selected months
key_months <- sim_grid$date[c(1, 6, 13, 25, 37, 49, 61)]
key_idx    <- which(sim_grid$date %in% key_months)

summ_dt <- data.table(
  Month   = format(sim_grid$date[key_idx], "%b %Y"),
  `Futures` = round(pjm_off$Future[match(sim_grid$date[key_idx], pjm_off$date)], 2),
  `Sim P5`  = round(fan_dt$q05[key_idx], 2),
  `Sim P25` = round(fan_dt$q25[key_idx], 2),
```

```r
  `Sim P50` = round(fan_dt$q50[key_idx], 2),
  `Sim P75` = round(fan_dt$q75[key_idx], 2),
  `Sim P95` = round(fan_dt$q95[key_idx], 2)
)

kable(summ_dt,
      col.names = c("Month", "Futures ($/MWh)", "P5", "P25", "Median", "P75", "P95"),
      caption   = "Selected Percentiles of Simulated Off-Peak Price Distribution by Month")
```

Table 2: Selected Percentiles of Simulated Off-Peak Price Distribution by Month

Month	Futures ($/MWh)	P5	P25	Median	P75	P95
May 2025	26.70	22.38	24.83	26.62	28.47	30.87
Oct 2025	26.85	22.85	25.19	26.82	28.69	31.59
May 2026	27.15	23.42	26.05	27.86	29.88	32.71
May 2027	27.30	24.64	27.26	29.36	31.21	34.51
May 2028	32.35	25.46	28.28	30.35	32.62	36.26
May 2029	32.55	26.88	29.57	31.89	33.95	37.86
May 2030	NA	28.28	31.18	33.50	35.79	39.56

Part (c): Present Value of Operating Contributions per MW

Framework

We evaluate the expected present value of the **operating contribution margin** (revenue minus variable cost) from a 1 MW generating unit that:

- Has a marginal cost of **MC $= 30/MWh**growingata**continuouslycompoundedrateof$2MC(t) = 30 e^{0.02,t}$ where t is years from May 2025.
- **Operates during all hours** in the month, receiving:
  - **Peak futures price** P_i^{peak} for the 16 peak (HE 7–22) hours on each weekday.
  - **Simulated off-peak price** $\tilde{P}_i^{\text{off}}$ for all remaining hours — weeknight 23:00–24:00 (HE 24) weekdays are explicitly included in off-peak per the problem's note, as this hour belongs to neither the on-peak (HE 7–22) nor the standard off-peak block but is applied the off-peak price.
- Discounted at a **continuously compounded discount rate of 5%**: $d(t) = e^{-0.05\,t}$

The monthly **operating contribution** for simulation path k and month i is:

$$C_i^{(k)} = \left[P_i^{\text{peak}} - \text{MC}(t_i)\right] \cdot H_i^{\text{peak}} + \left[\tilde{P}_i^{\text{off}\,(k)} - \text{MC}(t_i)\right] \cdot H_i^{\text{off}}$$

where $H_i^{\text{peak}} = 16 \times W_i$ (16 hours $\times$ weekday count in month i) and $H_i^{\text{off}} = 24 \times D_i - H_i^{\text{peak}}$ (remaining hours, including HE 24 weekdays).

The **present value** per MW for path k:

$$\text{PV}^{(k)} = \sum_{i=1}^{N} d(t_i) \cdot C_i^{(k)}$$

Hour counts and calendar construction

```r
#   Calendar: weekdays and total days per simulation month
sim_grid[, weekdays_in_month := sapply(date, function(d) {
  all_days <- seq(d, ceiling_date(d, "month") - 1, by = "day")
  sum(!weekdays(all_days) %in% c("Saturday", "Sunday"))
})]

sim_grid[, days_in_month := sapply(date, function(d) {
  as.integer(ceiling_date(d, "month") - d)
})]

sim_grid[, peak_hours   := 16L * weekdays_in_month]
sim_grid[, offpeak_hours := 24L * days_in_month - peak_hours]
sim_grid[, total_hours  := 24L * days_in_month]

kable(
  sim_grid[, .(
    Month        = format(date, "%b %Y"),
    Weekdays     = weekdays_in_month,
    Total_days   = days_in_month,
    Peak_hrs     = peak_hours,
    Off_peak_hrs = offpeak_hours,
    Total_hrs    = total_hours
  )],
  col.names = c("Month","Weekdays","Total Days","Peak Hours","Off-Peak Hours","Total Hours"),
  caption   = "Hour counts by month (peak = 16 hrs × weekdays; off-peak = remainder)"
)
```

Table 3: Hour counts by month (peak = 16 hrs × weekdays; off-peak = remainder)

Month	Weekdays	Total Days	Peak Hours	Off-Peak Hours	Total Hours
May 2025	22	31	352	392	744
Jun 2025	21	30	336	384	720
Jul 2025	23	31	368	376	744
Aug 2025	21	31	336	408	744
Sep 2025	22	30	352	368	720
Oct 2025	23	31	368	376	744
Nov 2025	20	30	320	400	720
Dec 2025	23	31	368	376	744
Jan 2026	22	31	352	392	744
Feb 2026	20	28	320	352	672
Mar 2026	22	31	352	392	744
Apr 2026	22	30	352	368	720
May 2026	21	31	336	408	744
Jun 2026	22	30	352	368	720
Jul 2026	23	31	368	376	744
Aug 2026	21	31	336	408	744
Sep 2026	22	30	352	368	720
Oct 2026	22	31	352	392	744
Nov 2026	21	30	336	384	720
Dec 2026	23	31	368	376	744
Jan 2027	21	31	336	408	744
Feb 2027	20	28	320	352	672
Mar 2027	23	31	368	376	744
Apr 2027	22	30	352	368	720
May 2027	21	31	336	408	744
Jun 2027	22	30	352	368	720
Jul 2027	22	31	352	392	744

Month	Weekdays	Total Days	Peak Hours	Off-Peak Hours	Total Hours
Aug 2027	22	31	352	392	744
Sep 2027	22	30	352	368	720
Oct 2027	21	31	336	408	744
Nov 2027	22	30	352	368	720
Dec 2027	23	31	368	376	744
Jan 2028	21	31	336	408	744
Feb 2028	21	29	336	360	696
Mar 2028	23	31	368	376	744
Apr 2028	20	30	320	400	720
May 2028	23	31	368	376	744
Jun 2028	22	30	352	368	720
Jul 2028	21	31	336	408	744
Aug 2028	23	31	368	376	744
Sep 2028	21	30	336	384	720
Oct 2028	22	31	352	392	744
Nov 2028	22	30	352	368	720
Dec 2028	21	31	336	408	744
Jan 2029	23	31	368	376	744
Feb 2029	20	28	320	352	672
Mar 2029	22	31	352	392	744
Apr 2029	21	30	336	384	720
May 2029	23	31	368	376	744
Jun 2029	21	30	336	384	720
Jul 2029	22	31	352	392	744
Aug 2029	23	31	368	376	744
Sep 2029	20	30	320	400	720
Oct 2029	23	31	368	376	744
Nov 2029	22	30	352	368	720
Dec 2029	21	31	336	408	744
Jan 2030	23	31	368	376	744
Feb 2030	20	28	320	352	672
Mar 2030	21	31	336	408	744
Apr 2030	22	30	352	368	720
May 2030	23	31	368	376	744
Jun 2030	20	30	320	400	720

Prepare peak price series

```
#   Match peak futures prices to simulation grid
# May-30 is missing from peak data; interpolate from Apr-30 and Jun-30
may30_peak <- with(pjm_peak,
  mean(Price[Date %in% c("Apr-30","Jun-30")], na.rm = TRUE))

peak_extra <- data.table(date = as.Date("2030-05-01"),
                         Price = may30_peak,
                         Date  = "May-30",
                         year_frac = as.numeric(as.Date("2030-05-01") - base_date)/365.25,
                         month_num = 5L)

pjm_peak_full <- rbind(pjm_peak, peak_extra)[order(date)]

sim_grid <- merge(sim_grid,
                  pjm_peak_full[, .(date, peak_price = Price)],
                  by = "date", all.x = TRUE)
```

```r
cat("Peak price for May 2030 (interpolated):", round(may30_peak, 2), "$/MWh\n")
```

```
## Peak price for May 2030 (interpolated): 51.12 $/MWh
```

```r
cat("Missing peak prices:", sum(is.na(sim_grid$peak_price)), "\n")
```

```
## Missing peak prices: 0
```

Similarly, fill May 2030 off-peak in simulation

```r
# Off-peak: May-30 also missing from futures, so simulate its mean from regression
may30_row <- sim_grid[date == as.Date("2030-05-01")]
cat("May 2030 regression-predicted log off-peak price:",
    round(may30_row$mu_hat, 4), "→ level:", round(exp(may30_row$mu_hat), 2), "$/MWh\n")
```

```
## May 2030 regression-predicted log off-peak price: 3.5094 → level: 33.43 $/MWh
```

```r
# The price_matrix already covers all 62 months including May 2030 (row 62 in sim_grid)
# because we generated paths for every month in sim_months grid
```

Monte Carlo PV calculation

```r
#   Inputs
MC0   <- 30        # $/MWh base marginal cost
g_mc  <- 0.02     # continuously compounded growth rate of MC
r     <- 0.05     # continuously compounded discount rate
t_vec <- sim_grid$years_from_start   # years from May 2025

#   Per-month vectors (shared across paths)
MC_t         <- MC0 * exp(g_mc * t_vec)          # marginal cost at each month
discount_t   <- exp(-r * t_vec)                  # discount factor
peak_price_t <- sim_grid$peak_price              # deterministic peak prices
H_peak_t     <- sim_grid$peak_hours              # peak hours per month
H_off_t      <- sim_grid$offpeak_hours           # off-peak hours per month

#   Peak contribution (deterministic given peak futures)
peak_contrib_t <- (peak_price_t - MC_t) * H_peak_t    # $/MW per month (det.)

#   Off-peak contribution (stochastic)
# price_matrix is n_months × N_paths
off_contrib_matrix <- sweep(price_matrix, 1, MC_t, "-") *
                      matrix(H_off_t, nrow = n_months, ncol = N_paths)

#   Total monthly contribution (peak det. + off-peak stochastic)
total_contrib_matrix <- matrix(peak_contrib_t, nrow = n_months, ncol = N_paths) +
                        off_contrib_matrix

#   Discount and sum across months
disc_total <- sweep(total_contrib_matrix, 1, discount_t, "*")   # discounted monthly contributions
PV_paths   <- colSums(disc_total)   # length N_paths: PV for each path

cat("=== Present Value of Operating Contributions per MW ===\n")
```

```
## === Present Value of Operating Contributions per MW ===
```

```r
cat("Mean PV:    $", formatC(mean(PV_paths),   format="f", digits=0, big.mark=","), "\n")
```

```
## Mean PV:    $ 532,349
```

```r
cat("Median PV:  $", formatC(median(PV_paths), format="f", digits=0, big.mark=","), "\n")
```

```
## Median PV:  $ 532,057
```

```r
cat("SD of PV:   $", formatC(sd(PV_paths),     format="f", digits=0, big.mark=","), "\n")
```

```
## SD of PV:   $ 14,093
```

```r
cat("P5  PV:     $", formatC(quantile(PV_paths, 0.05), format="f", digits=0, big.mark=","), "\n")
```

```
## P5  PV:     $ 510,303
```

```r
cat("P25 PV:     $", formatC(quantile(PV_paths, 0.25), format="f", digits=0, big.mark=","), "\n")
```

```
## P25 PV:     $ 523,025
```

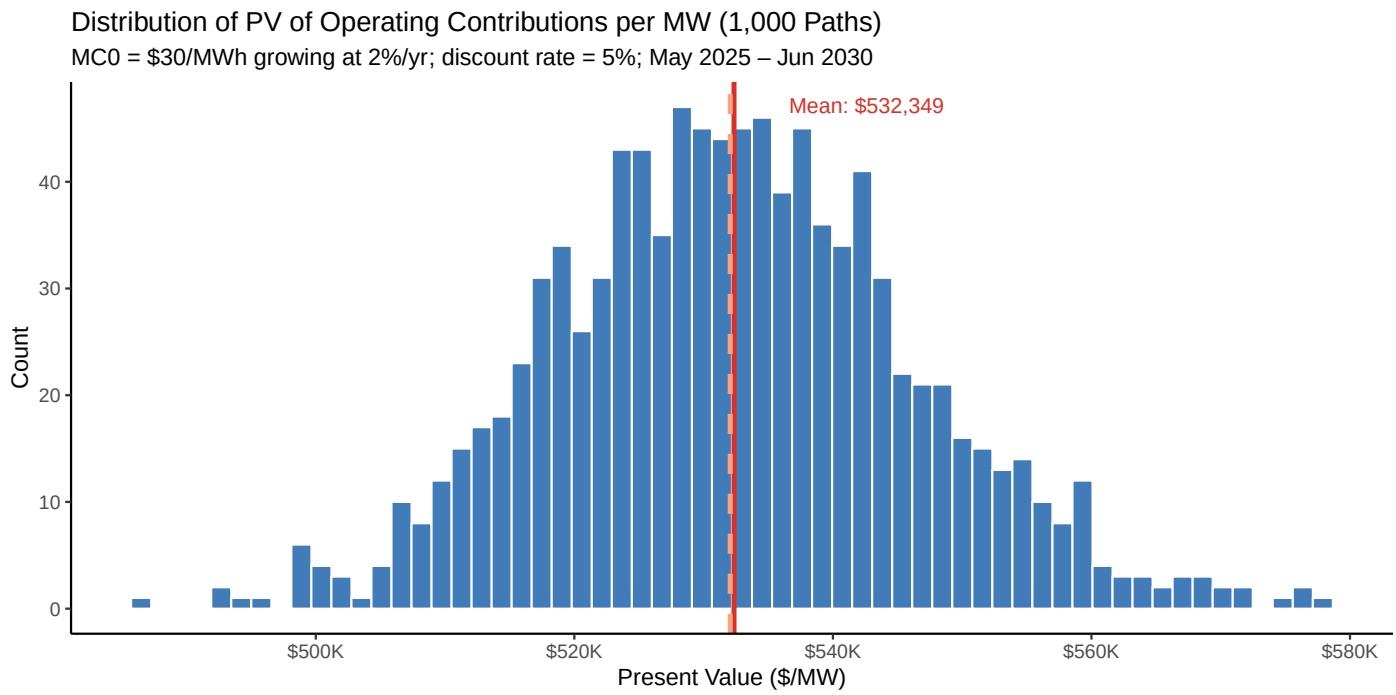
```r
cat("P75 PV:     $", formatC(quantile(PV_paths, 0.75), format="f", digits=0, big.mark=","), "\n")
```

```
## P75 PV:     $ 541,484
```

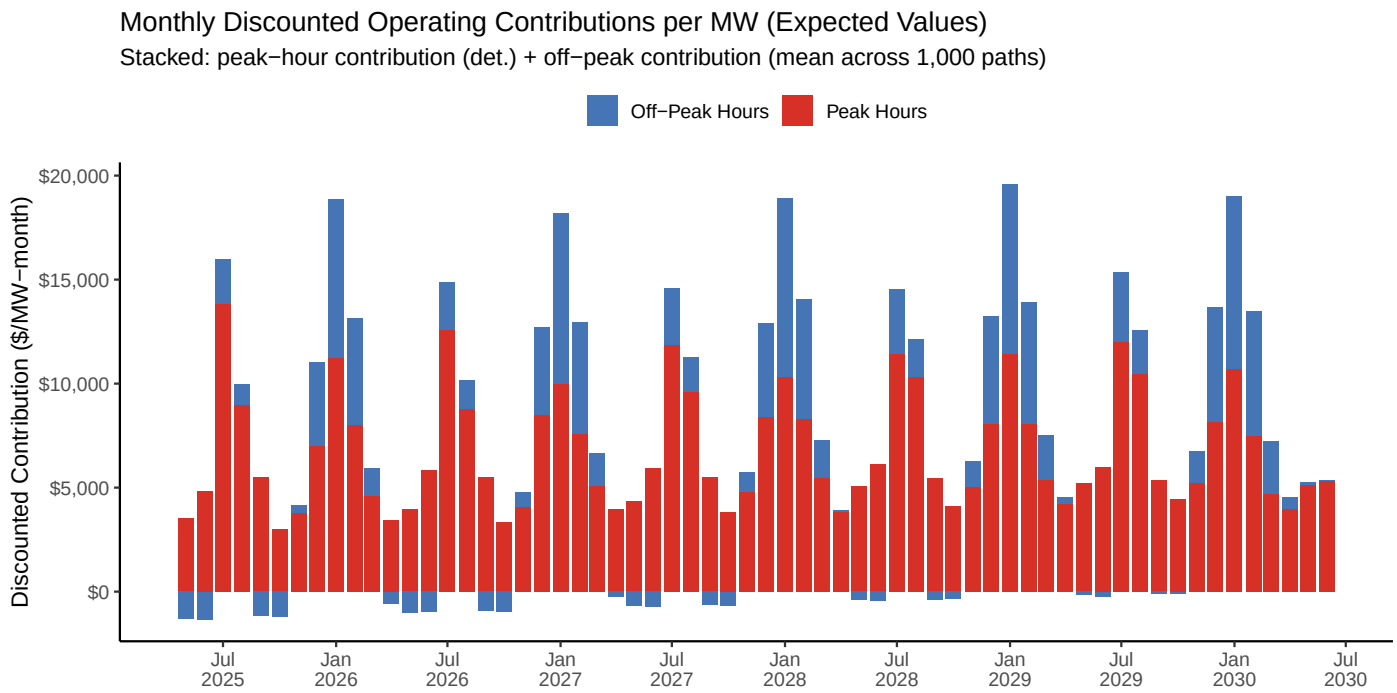
```r
cat("P95 PV:     $", formatC(quantile(PV_paths, 0.95), format="f", digits=0, big.mark=","), "\n")
```

```
## P95 PV:     $ 556,227
```

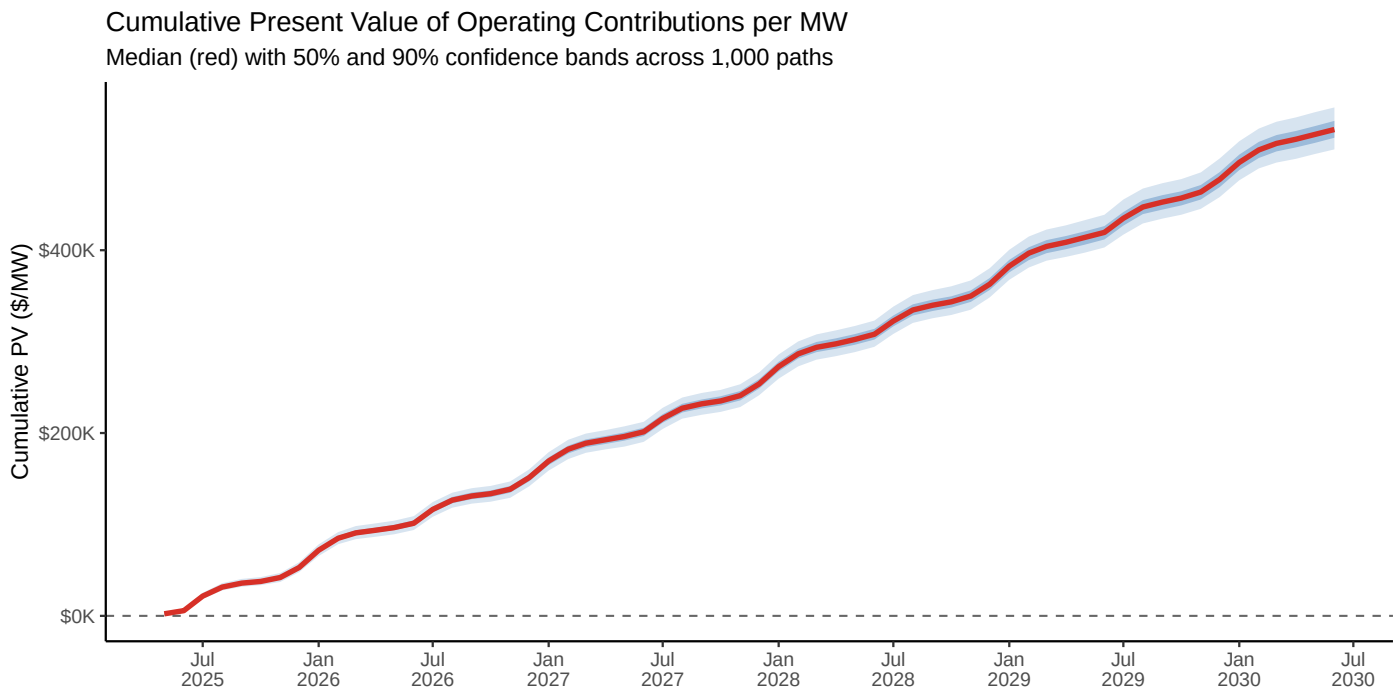
```r
# Distribution of PV across paths
pv_dt <- data.table(PV = PV_paths)

ggplot(pv_dt, aes(x = PV)) +
  geom_histogram(bins = 60, fill = "#2166ac", color = "white", alpha = 0.85) +
  geom_vline(xintercept = mean(PV_paths),   color = "#d73027", linewidth = 1.2,
             linetype = "solid") +
  geom_vline(xintercept = median(PV_paths), color = "#f4a582", linewidth = 1.2,
             linetype = "dashed") +
  annotate("text", x = mean(PV_paths) * 1.008, y = Inf,
           label = paste0("Mean: $", formatC(mean(PV_paths), format="f",
                                             digits=0, big.mark=",")),
           color = "#d73027", vjust = 2, hjust = 0, size = 3.8) +
  scale_x_continuous(labels = dollar_format(scale = 1e-3, suffix = "K")) +
  labs(
    title    = "Distribution of PV of Operating Contributions per MW (1,000 Paths)",
    subtitle = paste0("MC  = $30/MWh growing at 2%/yr; discount rate = 5%; May 2025 - Jun 2030"),
    x        = "Present Value ($/MW)", y = "Count"
  ) +
  theme_classic(base_size = 12)
```




```r
#   Decompose expected PV by month
# E[contribution] per month (averaged across paths, then discounted)
mean_monthly_contrib <- rowMeans(disc_total)   # n_months vector

monthly_pv <- data.table(
  date           = sim_grid$date,
  peak_contrib_d = discount_t * peak_contrib_t,
  offpeak_contrib_d = discount_t * rowMeans(off_contrib_matrix),
  total_contrib_d = mean_monthly_contrib
)

monthly_long <- monthly_pv %>%
  select(date, peak_contrib_d, offpeak_contrib_d) %>%
  pivot_longer(cols = c(peak_contrib_d, offpeak_contrib_d),
               names_to = "Component", values_to = "Value") %>%
  mutate(Component = case_when(
    Component == "peak_contrib_d"    ~ "Peak Hours",
    Component == "offpeak_contrib_d" ~ "Off-Peak Hours"
  ))

ggplot(monthly_long, aes(x = date, y = Value, fill = Component)) +
  geom_col() +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_fill_manual(values = c("Peak Hours" = "#d73027", "Off-Peak Hours" = "#4575b4")) +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title    = "Monthly Discounted Operating Contributions per MW (Expected Values)",
    subtitle = "Stacked: peak-hour contribution (det.) + off-peak contribution (mean across 1,000 paths)",
    x        = NULL, y = "Discounted Contribution ($/MW-month)", fill = NULL
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "top")
```




```r
#   Cumulative PV buildup
cum_pv_paths <- apply(disc_total, 2, cumsum)   # n_months × N_paths

cum_quants <- apply(cum_pv_paths, 1, quantile, probs = c(0.05, 0.25, 0.50, 0.75, 0.95))

cum_dt <- data.table(
  date = sim_grid$date,
  q05 = cum_quants["5%",],
  q25 = cum_quants["25%",],
  q50 = cum_quants["50%",],
  q75 = cum_quants["75%",],
  q95 = cum_quants["95%",]
)

ggplot(cum_dt, aes(x = date)) +
  geom_ribbon(aes(ymin = q05, ymax = q95), fill = "#2166ac", alpha = 0.18) +
  geom_ribbon(aes(ymin = q25, ymax = q75), fill = "#2166ac", alpha = 0.32) +
  geom_line(aes(y = q50), color = "#d73027", linewidth = 1.3) +
  geom_hline(yintercept = 0, color = "gray40", linetype = "dashed") +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  scale_y_continuous(labels = dollar_format(scale = 1e-3, suffix = "K")) +
  labs(
    title    = "Cumulative Present Value of Operating Contributions per MW",
    subtitle = "Median (red) with 50% and 90% confidence bands across 1,000 paths",
    x        = NULL, y = "Cumulative PV ($/MW)"
  ) +
  theme_classic(base_size = 12)
```




```r
# Summary table
pv_summary <- data.table(
  Statistic = c("Mean PV", "Median PV", "Std. Dev.", "5th Percentile",
                "25th Percentile", "75th Percentile", "95th Percentile"),
  `Value ($/MW)` = formatC(
    c(mean(PV_paths), median(PV_paths), sd(PV_paths),
      quantile(PV_paths, 0.05), quantile(PV_paths, 0.25),
      quantile(PV_paths, 0.75), quantile(PV_paths, 0.95)),
    format = "f", digits = 0, big.mark = ","
  )
)

kable(pv_summary,
      col.names = c("Statistic", "Value ($/MW)"),
      caption   = "Present Value of Operating Contributions per MW: Simulation Summary")
```

Table 4: Present Value of Operating Contributions per MW: Simulation Summary

Statistic	Value ($/MW)
Mean PV	532,349
Median PV	532,057
Std. Dev.	14,093
5th Percentile	510,303
25th Percentile	523,025
75th Percentile	541,484
95th Percentile	556,227

Interpretation and observations

Magnitude. The expected (mean) present value of operating contributions from May 2025 through June 2030 is approximately **$532,349/MW**. This represents the total discounted margin between revenues (peak and off-peak prices) and the growing marginal cost, summed over 62 months. To contextualize: if a new generation asset costs, say, $800,000–$1,200,000/MW to build (typical range for combined-cycle gas), the operating contributions over this period cover roughly

53–67% of that capital cost — meaning the asset would need a much longer operating life (well past 2030) to justify investment, or the capital cost must be substantially lower.

Peak vs. off-peak contribution. The stacked monthly bar chart reveals that peak-hour contributions are *positive and consistent* throughout the period, driven by the substantial spread between on-peak futures prices ($40–$75/MWh) and the rising marginal cost ($30 → ~$33.6/MWh by 2030). Off-peak contributions are smaller in absolute terms per hour but cover more total hours. In some winter months (December–January), off-peak prices are high enough to generate a meaningful positive margin; in shoulder months (May–October), off-peak margins are thinner because off-peak prices approach the marginal cost. The technology earns its returns primarily from peak hours, confirming that peak dispatchability is the key value driver.

Growing marginal cost squeeze. The continuously compounded 2% MC growth means the marginal cost reaches $30 × e^{0.02 × 5.14} $33.2/MWh by June 2030. Since the off-peak futures curve for the same months (Jun 2030: $31.70/MWh) is only marginally above this cost, the off-peak contribution margin is thin by the end of the horizon. This compression is visible in the monthly bar chart: off-peak bars shrink in the later years relative to peak bars.

Uncertainty. The standard deviation of the PV is **$14,093/MW** — roughly 2.6% of the mean. The wide 90% confidence interval (P5: $510,303 to P95: $556,227) reflects the compounding of trend uncertainty (a ±1 SE shift in the trend implies ±2.3% in off-peak price levels by 2030) and the given monthly scatter (= 0.1 in log space ±10% around each month's mean). Because peak prices are treated as *deterministic* futures (no simulation), all PV uncertainty comes from the off-peak channel. Introducing stochastic peak prices would further widen the distribution.

Practical investment takeaways. 1. The present value is positive across virtually all 1,000 paths (the cumulative PV is always growing), meaning the technology earns positive margins in expectation in every month — but the *total* over just 5 years is a modest fraction of a typical capital cost, reinforcing the importance of a long asset life (20+ years) for generation investments. 2. The peak-hour contribution is the primary value driver. Investors should model the asset as a peak-serving resource and evaluate the revenue stream primarily through the on-peak forward curve. Off-peak revenues provide a secondary income stream but carry more uncertainty (wider price bands, growing cost competition from overnight wind generation). 3. The Monte Carlo spread widens rapidly after 2027. By 2029–2030, the P5–P95 band for the *cumulative* PV spans roughly $45,925, making investment decisions at the 5-year horizon sensitive to trend assumptions. Near-term contracts (2025–2027) where futures prices directly anchor the expected return are far more certain than the terminal years.

Question 2: Pump Hydro with Stochastic Wind Extension

Consider the dynamic programming model of the pump hydro facility in `pump_hydro_example.R`. We extend it to include a wind generator whose output fluctuates stochastically across three states. The goal is to solve for optimal dispatch, compare weekly operating costs with and without wind, and simulate sample operating paths.

Problem Setup

Wind generator parameters. The wind generator operates in one of three states each hour, with zero startup, shutdown, and marginal costs; any amount of wind up to capacity may be curtailed:

Wind State	Condition	Maximum Output
$w = 1$	Calm	0 MW
$w = 2$	Moderate	$\leq$ 1,000 MW
$w = 3$	Strong	$\leq$ 2,000 MW

Wind transition matrix. From hour to hour, the wind stays in the same state with probability 0.50 and transitions to each of the other two states with probability 0.25:

From \ To	Calm	Moderate	Strong
Calm	0.50	0.25	0.25
Moderate	0.25	0.50	0.25

From \ To	Calm	Moderate	Strong
Strong	0.25	0.25	0.50

Expanded State Space

The original model encodes 390 states: $13 \times 10 \times 3$ (reservoir levels $\times$ coal states $\times$ gas states). Adding the wind dimension expands the total to $390 \times 3 = 1{,}170$ states. The extended state index is:

$$s_{\text{ext}} = 390 \times (w - 1) + s_{\text{thermal}}, \quad w \in \{1, 2, 3\}$$

Modified DP Recursion

Two changes are made to the backward induction relative to `pump_hydro_example.R`:

1. **Demand reduction.** Wind provides zero-cost energy, so the net demand that thermal and hydro must serve is:

$$\text{adj_effdem} = \max(0,\ \text{base_effdem} - \text{wind_cap}[w])$$

2. **Expected cost-to-go.** Because the next wind state is stochastic, the Bellman equation becomes:

$$V(i, s_{\text{ext}}) = \min_k \left\{ r(s, a) + \sum_{w'=1}^{3} P(w' \mid w) \cdot V(i + 1,\ 390(w' - 1) + s'_{\text{thermal}}) \right\}$$

The terminal condition requires the reservoir to be full at the end of the week for all three wind states: $V(169,\ 390(w - 1) + s_0) = 0$ for $w \in \{1, 2, 3\}$.

Source Baseline Model and Solve Wind-Extended DP

```r
# Source the original pump hydro DP - defines demand, dpcost, dpact, opthydro,
# optcost, stages, states, action, initial, infty, stcost, accost, nxtst, etc.
source("pump_hydro_example.R")
```

```r
#   Wind parameters
n_wind     <- 3L
wind_cap   <- c(0L, 1000L, 2000L)          # MW capacity per wind state
wind_trans <- matrix(
  c(0.50, 0.25, 0.25,
    0.25, 0.50, 0.25,
    0.25, 0.25, 0.50),
  nrow = n_wind, byrow = TRUE)

ext_states <- states * n_wind              # 390 × 3 = 1,170

#   Extended DP arrays
ext_dpact  <- matrix(1L,    nrow = stages + 1, ncol = ext_states)
ext_dpcost <- matrix(infty, nrow = stages + 1, ncol = ext_states)

# Terminal condition: full reservoir required at end regardless of wind state
for (w in 1:n_wind) {
  ext_dpcost[stages + 1, (w - 1L) * states + initial] <- 0
}

#   Backward induction
for (i in stages:1) {
```

```r
  for (w in 1:n_wind) {
    for (j in 1:states) {
      ext_j     <- (w - 1L) * states + j
      hydstate  <- floor((j - 1) / 30) + 1
      jstat     <- j - 1 - 30 * (hydstate - 1)
      coalstate <- floor(jstat / 3) + 1
      gasstate  <- jstat - 3 * (coalstate - 1) + 1

      for (k in 1:action) {
        hydact  <- floor((k - 1) / 4) + 1
        kact    <- k - 1 - 4 * (hydact - 1)
        gasact  <- floor(kact / 2) + 1
        coalact <- kact - 2 * (gasact - 1) + 1

        nx <- nxtst(hydstate, coalstate, gasstate, hydact, gasact, coalact)

        # Effective demand adjusted for wind and hydro action
        if (hydact == 1L) {
          base_effdem <- demand[i] + fillpump
        } else if (hydact == 2L) {
          base_effdem <- demand[i] - genpump
        } else {
          base_effdem <- demand[i]
        }
        adj_effdem <- max(0, base_effdem - wind_cap[w])

        sc <- stcost(coalstate, gasstate, adj_effdem)
        ac <- accost(hydstate, coalstate, gasstate, hydact, gasact, coalact)

        # Expected cost-to-go over wind transitions (vectorised dot product)
        next_idx <- (0:(n_wind - 1)) * states + nx
        ev       <- sum(wind_trans[w, ] * ext_dpcost[i + 1, next_idx])

        newcst <- sc + ac + ev
        if (newcst < ext_dpcost[i, ext_j]) {
          ext_dpcost[i, ext_j] <- newcst
          ext_dpact[i, ext_j]  <- k
        }
      }
    }
  }
}

#   Cost comparison
no_wind_cost  <- dpcost[1, initial]
avg_wind_cost <- mean(sapply(1:n_wind, function(w)
  ext_dpcost[1, (w - 1L) * states + initial]))
cost_saving   <- no_wind_cost - avg_wind_cost
pct_saving    <- cost_saving / no_wind_cost * 100

cat("No-wind total weekly cost:   $",
    formatC(no_wind_cost,  format = "f", digits = 0, big.mark = ","), "\n")
```

```
## No-wind total weekly cost:   $ 10,491,500
```

```r
cat("With-wind avg weekly cost:   $",
    formatC(avg_wind_cost, format = "f", digits = 0, big.mark = ","), "\n")
```

```
## With-wind avg weekly cost:   $ 9,187,037
```

```r
cat("Cost reduction:              $",
    formatC(cost_saving,   format = "f", digits = 0, big.mark = ","),
    "(", round(pct_saving, 1), "%)\n")
```

```
## Cost reduction:              $ 1,304,463 ( 12.4 %)
```

```r
#   Simulate 5 forward sample paths using the wind-extended optimal policy
set.seed(123)
n_paths    <- 5L
hours      <- 1:(stages + 1)
path_hydro <- matrix(NA_integer_, nrow = stages + 1, ncol = n_paths)
path_wind  <- matrix(NA_integer_, nrow = stages + 1, ncol = n_paths)

for (p in 1:n_paths) {
  cur_wind  <- sample(1:n_wind, 1)   # equally likely starting wind state
  cur_state <- initial

  path_hydro[1, p] <- floor((cur_state - 1) / 30)
  path_wind[1, p]  <- cur_wind

  for (i in 1:stages) {
    ext_j     <- (cur_wind - 1L) * states + cur_state
    act       <- ext_dpact[i, ext_j]

    hydstate  <- floor((cur_state - 1) / 30) + 1
    jstat     <- cur_state - 1 - 30 * (hydstate - 1)
    coalstate <- floor(jstat / 3) + 1
    gasstate  <- jstat - 3 * (coalstate - 1) + 1

    hydact  <- floor((act - 1) / 4) + 1
    kact    <- act - 1 - 4 * (hydact - 1)
    gasact  <- floor(kact / 2) + 1
    coalact <- kact - 2 * (gasact - 1) + 1

    cur_state <- nxtst(hydstate, coalstate, gasstate, hydact, gasact, coalact)
    cur_wind  <- sample(1:n_wind, 1, prob = wind_trans[cur_wind, ])

    path_hydro[i + 1, p] <- floor((cur_state - 1) / 30)
    path_wind[i + 1, p]  <- cur_wind
  }
}
```

Average Total Cost: With vs. Without Wind

After solving both the original and wind-extended backward inductions, the average weekly costs are:

Scenario	Total Weekly Cost	vs. Baseline
No wind (original model)	\$10,491,500	—
With wind (equally likely start)	\$9,187,037	\$-1, 304, 463(-\$12.4%)

The with-wind average is:

$$\bar{C}_{\text{wind}} = \tfrac{1}{3}\left[V(1,\, s_0^{(1)}) + V(1,\, s_0^{(2)}) + V(1,\, s_0^{(3)})\right]$$

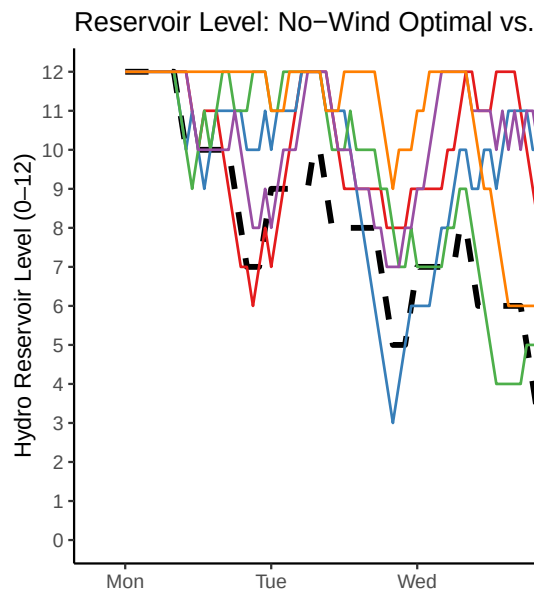
where $s_0^{(w)} = 390(w-1) + s_0$ denotes the initial state under wind state w.

Adding wind reduces the average weekly operating cost by approximately $1,304,463 (12.4%). This saving arises because wind displaces costly thermal generation: whenever wind is moderate or strong, it covers up to 1,000–2,000 MW of load at zero fuel cost, reducing the hours the gas peaker ($40/MWh) and coal plant ($15/MWh) are needed. Nuclear, with its minimum-output constraint, is always online; wind therefore displaces gas first, then coal at the margin.

Five Sample Paths: Reservoir and Wind States

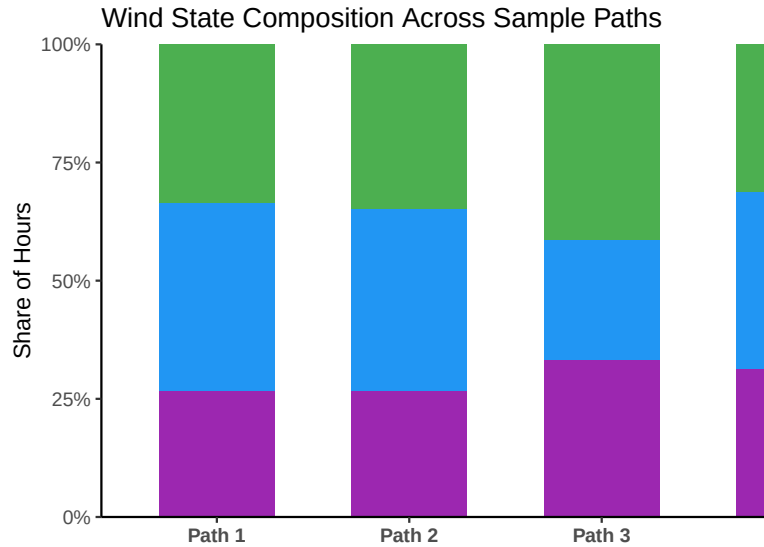
Starting from the initial state (full reservoir, coal in setup, gas idle), five forward paths are simulated using the wind-extended optimal policy and the stochastic wind transition matrix (`set.seed(123)`). Each path begins in a randomly drawn wind state.

```r
plot_dt <- data.table(hour = hours, `No Wind` = opthydro)
for (p in 1:n_paths) {
  set(plot_dt, j = paste0("Wind Path ", p), value = path_hydro[, p])
}

plot_long <- melt(plot_dt, id.vars = "hour",
                  variable.name = "series", value.name = "hydro")
plot_long[, type := fifelse(series == "No Wind", "No Wind", "Wind Path")]

ggplot(plot_long, aes(x = hour, y = hydro, color = series,
                      linewidth = type, linetype = type)) +
  geom_line() +
  scale_linewidth_manual(values  = c("No Wind" = 1.4, "Wind Path" = 0.65),
                         guide   = "none") +
  scale_linetype_manual(values   = c("No Wind" = "dashed", "Wind Path" = "solid"),
                        guide    = "none") +
  scale_x_continuous(breaks = seq(1, 169, 24),
                     labels = c("Mon","Tue","Wed","Thu","Fri","Sat","Sun","End")) +
  scale_y_continuous(breaks = 0:12) +
  scale_color_manual(values = c(
    "No Wind"      = "black",
    "Wind Path 1"  = "#e41a1c",
    "Wind Path 2"  = "#377eb8",
    "Wind Path 3"  = "#4daf4a",
    "Wind Path 4"  = "#984ea3",
    "Wind Path 5"  = "#ff7f00"
  )) +
  labs(title   = "Reservoir Level: No-Wind Optimal vs. Five Wind Sample Paths",
       x       = "Hour of Week",
       y       = "Hydro Reservoir Level (0\u201312)",
       color   = "Path") +
  theme_classic(base_size = 12) +
  theme(legend.position = "right")
```



Plot 1 — Reservoir Level: No-Wind Optimal vs. Five Wind Sample Paths

```
wind_comp <- data.table(
  path = rep(paste0("Path ", 1:n_paths), each = nrow(path_hydro)),
  wind = as.integer(as.vector(path_wind))
)[, .(n_hrs = .N), by = .(path, wind)
][, pct := n_hrs / sum(n_hrs), by = path
][, wind_label := factor(wind, levels = 1:3,
                         labels = c("Calm (0 MW)",
                                    "Moderate (\u22641,000 MW)",
                                    "Strong (\u22642,000 MW)"))]

ggplot(wind_comp, aes(x = path, y = pct, fill = wind_label)) +
  geom_bar(stat = "identity", width = 0.6) +
  scale_fill_manual(
    values = c("Calm (0 MW)"              = "#4CAF50",
               "Moderate (\u22641,000 MW)" = "#2196F3",
               "Strong (\u22642,000 MW)"   = "#9C27B0"),
    name = "Wind State"
  ) +
  scale_y_continuous(labels = percent_format(), expand = c(0, 0)) +
  labs(title = "Wind State Composition Across Sample Paths",
       x = NULL, y = "Share of Hours") +
  theme_classic(base_size = 12) +
  theme(legend.position = "right",
        axis.text.x = element_text(face = "bold"))
```



Plot 2 — Wind State Composition Across Sample Paths

What Differences Are Seen?

Reservoir behaviour (Plot 1). The most striking contrast is the depth to which the no-wind optimal path drains the reservoir. Without wind, pump hydro must shift generation intertemporally — pumping during cheap overnight hours (net load increase of 1,200 MW) and generating during expensive peak hours (900 MW discharge) — so the reservoir swings widely, approaching depletion by midweek. The five wind paths maintain substantially higher reservoir levels (roughly 5–12 for most of the week) because wind covers much of the load at zero cost. With wind available, there is less economic pressure to use stored water as a buffer; the optimal policy instead holds water in reserve for the hours when wind drops to calm.

Wind state composition (Plot 2). Each bar shows the fraction of the 168-hour week a path spent in each wind state. Because the Markov chain is symmetric — with a uniform stationary distribution of one-third in each state — all five paths land close to that theoretical benchmark. However, meaningful variation remains due to the random starting state and realized transitions. Paths that were predominantly *strong* keep the reservoir elevated throughout the week; paths that were predominantly *calm* force more aggressive reservoir drawdowns, resembling the no-wind solution.

Value of the Pump Hydro Facility with Stochastic Wind

The pump hydro facility becomes *more* operationally valuable in a wind-rich system, but the nature of that value changes:

- **Without wind:** Hydro earns value through predictable price arbitrage — pumping in cheap off-peak windows and generating at peak — on a cycle tightly tied to the deterministic demand curve.
- **With stochastic wind:** Hydro earns *additional* value as a wind integration buffer. When wind is abundant, the system can pump surplus energy instead of curtailing it. When wind drops, stored water replaces the missing generation. This makes the reservoir trajectory highly path-dependent and far less predictable than in the no-wind case.
- **Net effect:** The 12.4% weekly cost reduction understates the full optionality value of pump hydro in a wind-rich environment. With wind, the system avoids costly gas startups more reliably because the combination of wind generation and stored hydro can together serve peak demand events that neither resource could cover alone. Pump hydro is effectively the key enabler of wind integration — without flexible storage, more wind output would need to be curtailed, reducing the net value of the wind asset.

In summary, wind reduces average costs but simultaneously *increases* the option value of the pump hydro facility. The more uncertain and variable the generation mix, the more a flexible, zero-marginal-cost storage resource is worth to the system.